Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/10/2024
PI Name:	Zenhausern, Frederic Ph.D.		
Project Title:	Space Biospecimen Collection and Accelerated Test for In-Flight Mission		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	
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City:	Phoeniz	**State:**	AZ
Zip Code:	85004	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2021	**End Date:**	04/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:	SpaceX Inspiration4 NOTE: Project title changed per TRISH information; former title listed in the NASA Task Book was "Miniaturized Inflight Blood/Saliva Analysis" (Ed., 4/10/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-EXP0006		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EXPAND Project Point-of-care (POC) assays are essential diagnostic platforms that allow fast health monitoring and disease testing with minimal equipment. On planet Earth, their use has shown to improve both clinical and economic outcome by, for example, allowing faster decision, starting treatment earlier, reducing use of staff, equipment, and hospital admissions. The likelihood for future long missions beyond low Earth orbit and planetary exploration makes a fast return on Earth difficult in case of medical emergency or punctual deterioration of health status. Therefore, there is a critical need to develop autonomous platforms providing a detailed medical diagnosis to help crew to take decision on self-therapeutic actions. Therefore, POC devices are perfectly suited for space missions that require equipment with reduced weight and size, minimal waste generation, fast and accurate data for rapid clinical management with limited staff and material resources. In this context, under Translational Research Institute for Space Health (TRISH) funding #NNX16AO69A SYN00-03&05, we developed an ultra-sensitive paper-based microfluidic Vertical Flow Immunoassay diagnostic platform (VFI), which can detect multiplex biomarkers ranging from “immuno-to-gene” signatures in biofluids to monitor health outcomes in-flight conditions during long-term space missions, ultimately. With the sudden COVID-19 pandemic, we rapidly adapted this VFI platform to an even more compact format that does not require any power supply or any heavy instrumental handling, and that can be performed without the need of medical facilities, in less than 1 hour, from sample collection to data analysis (TRISH #CAT0001). The goal of the present project is to validate this simplified version of the VFI in a microgravity environment during the Inspiration4 mission, a flight to low Earth orbit operated by SpaceX. Crew members will collect a small volume of whole blood from a finger prick and plasma will be isolated using a POC microfluidic cartridge for preparing proteins to be detected quickly with the VFI device before imaging the results with a software installed onto i-Phone or i-Pad devices. Saliva and dried blood spots (DBS) samples will also be collected, in addition to the pre- and post-flight samples, (1) to identify the immunological changes and potential (2) viral activation that could happen during short space flights, and (3) to supply samples for the new SpaceX/BCM biobank.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The benefits of the research projects comprised: 1) Validation of the design and prototyping of the vertical flow assay platform technology for multiplex molecular diagnostics for space related health applications. 2) Demonstration of scalability for production in compliance with the requirements of space flight missions improving Technology Readiness Level (TRL) maturity. 3) VFI user interface was successfully demonstrated which can be extended for other applications on Planet Earth, i.e. for home use. 4) The mobility and versatility of biosignatures integration opened the domain of applications of the VFI technology platform across federal agencies from NASA to Department of Defense (DoD), Biomedical Advanced Research and Development Authority (BARDA), and National Institutes of Health (NIH) ranging in applications from pathogens detection to biodosimetry of relevance to broad medical countermeasures.
Task Progress:	The main findings comprise knowledge gained in the operations of a space mission, compliance requirements for the technology development and scientific results. Below are some of the main findings: 1) During a short period, our team designed, prototyped and tested the proposed simple VFI device while training the sponsor and crew members how to use the devices within the environmental conditions of the spacecraft. This knowledge assisted in the understanding of the space operability of the first molecular diagnostic platform run by civilian crew members during a multiple days flight mission. 2) The device integrated multiple fluidic components allowing reagent solutions to be preloaded in sponge-based and 3D-printed plastic cartridges and stored in nonpermeable tubes and capsules. A pump-less microfluidic device was used to separate plasma and transfer it to VFI. The multiplexing capability was evaluated by simultaneous detection of C-Reactive Protein (CRP) and total immunoglobulin M (IgM) as stress, cardiovascular and inflammatory biomarker models. The major outcome of these in-flight experiments revealed efficient user-interface and VFI ability to operate in microgravity environment. 3) Virome analysis revealed that no major risk of any re-activation of most of the human known viral infections were detected amongst the crew. Only their recent vaccination status or exposure to SARS-COV2 were measured. 3) Long term shelf life of reagents has been determined as the area for improvement which has been initiated benefiting from the report of information post-flight.
Bibliography Type:	Description: (Last Updated: 04/20/2026)
Abstracts for Journals and Proceedings	Lacombe J, Summers AJ, Devadhasan J, Gu J, Nordquist A, Smith S, Thomas B, Altin J, Mead H, Nelson G, Kelley E, Mateus J, Banerjee NT, Menon AS, Zenhausern F. "Health molecular testing for commercial space mission." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
Articles in Peer-reviewed Journals	Jones CW, Overbey EG, Lacombe J, Ecker AJ, Meydan C, Ryon K, Tierney B, Damle N, MacKay M, Afshin EE, Foox J, Park J, Nelson T, Suhail M, Byhaqui SG, Aslam B, Tali UA, Nisa L, Menon P, Patel CO, Khan SA, Ebert DJ, Everson A, Schubert MC, Ali NN, Sarma MS, Kim J, Houerbi N, Grigorev K, Garcia Medina S, Summers AJ, Gu J, Altin JA, Fattahi A, Hirzallah MI, Wu JH, Stahn AC, Beheshti A, Klotz R, Ortiz V, Yu M, Patras L, Matei I, Lyden D, Melnick A, Banerjee N, Mullane S, Kleinman A, Loesche M, Menon AS, Donoviel DB, Urquieta E, Mateus J, Sargsyan AE, Shelhamer M, Zenhausern F, Bershad EM, Basner M, Mason CE. "Molecular and physiological changes in the SpaceX Inspiration4 civilian crew." Nature. 2024 Jun 11. https://doi.org/10.1038/s41586-024-07648-x ; PMID: 38862026 , Jun-2024

Significant Media Coverage	Smith B. (Zenhausern F interview). "UArizona research project to monitor health of SpaceX Inspiration4 crew members during mission. Dr. Zenhausern interviewed for university publication." August 30, 2021. https://phoenixmed.arizona.edu/spacex#:~:text=Researchers%20with%20the%20Center%20for ,gravity%20environment%20to%20monitor%20the , Aug-2021